

Numerical Analysis — Homework 5

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1. Let us apply Newton type method for solving the unconstrained optimization problem

$$\min_x F(x) \quad \text{for} \quad F(x) = \frac{1}{4}x_1^4 + x_1x_2 + \frac{1}{4}x_2^4.$$

- What is the gradient of the function F at a given point x ?
- What is the Hessian matrix of F at a given point x ?
- We implement a Newton type method method of the form

$$x^{k+1} = x^k - M^{-1} \nabla F(x_k)$$

with the constant Hessian approximation

$$M = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

What is the local convergence rate of this method?

2. Find the explicit solution $x(t)$ of the following scalar differential equations:

- $\dot{x}(t) = 2$ with $x(0) = -1$,
- $\dot{x}(t) = -x(t)$ with $x(0) = 1$,
- $\dot{x}(t) = -x(t) + 1$ with $x(0) = 0$,
- $\dot{x}(t) = -\frac{1}{x} e^{x^2}$ with $x(0) = 1$,
- $\dot{x}(t) = -\sin(t)$ with $x(0) = 1$.

3. Implement the standard Runge-Kutta method of order 4 with constant step size h for the differential equation

$$\forall t \in [0, 10], \quad \left\{ \begin{array}{l} \dot{x}_1(t) = -x_1(t) + \frac{1}{10}x_1(t)x_2(t) \\ \dot{x}_2(t) = -x_2(t) - x_2(t)^3 \end{array} \right\} \quad \text{with} \quad x_1(0) = x_2(0) = 1.$$

for $h = 0.1$, $h = 0.01$ and $h = 0.001$. Plot your results. Can you estimate the numerical approximation error, $\max_k \|y_k - x(t_k)\|_\infty$ for $0 \leq t_k = 0.01 * k \leq 10$, of the approximate solution $(y_0, y_1, \dots, y_{1000})$ for $h = 0.01$ by using your result for the step size $h = 0.001$? Justify your arguments.